Akash Kumar

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SKILLS

Languages: C++, Java, JavaScript
Frameworks: HTML, CSS, React.js, Next.js, Node.js, Express.js
Databases: MySQL, MongoDB
Tools/Platform: Git, GitHub, Postman, Docker
Soft Skills: Problem-Solving, Leadership, Team Collaboration, Adaptablity

PROJECTS

Context Tube | GitHub | Livelink | *Next.js, PostgreSQL, Drizzle ORM, Upstash, Google GenAI SDK, Docker*

- Developed an AI-powered Retrieval-Augmented Generation (RAG) platform that processes YouTube videos and enables interactive question answering over video content.
- Architected semantic search using PostgreSQL with pgvector, storing 768-dimensional embeddings generated through Google's Gemini API..
- Designed an asynchronous processing pipeline using Upstash Workflow and QStash for transcript processing, translation, chunking, and vectorization.
- Implemented retry handling with exponential backoff and ScrapeOps proxy aggregation to handle Gemini API rate limits and YouTube IP restrictions.
- Containerized the application using multi-stage Next.js Docker builds, reducing the deployment image from approximately 1GB to 120MB.

Dev Match | GitHub | Livelink | *React.js, Node.js, Express.js, MongoDB, Redis, Socket.io, Cloudinary, Resend*

- Built a developer networking platform featuring a Tinder-style discovery feed and bidirectional connection system with Redux Toolkit-based state management.
- Built a real-time chat system using Socket.io with optimistic UI updates, using cursor-based pagination and reverse infinite scrolling for efficient message loading.
- Designed scalable security and media pipelines with direct-to-cloud uploads using Cloudinary, Redis-backed rate limiting, OTP-based authentication, and OAuth 2.0 authentication (Google/GitHub).
- Integrated Razorpay payments with webhook signature verification for premium membership provisioning and access control.

Digital Coupon Manager | GitHub | Livelink | *React.js,· Node.js, Express.js, MongoDB, Tailwind CSS , JWT*

- Created an automated background service using Nodemailer to trigger real-time expiration alerts, effectively mitigating user asset loss through timely notifications.
- Engineered a responsive frontend with advanced filtering and sorting algorithms, optimizing the retrieval of digital assets and improving application accessibility and usability.
- Designed and implemented a secure authentication and authorization system using JWT, ensuring protected access to user-specific coupon data while maintaining scalability and enhancing overall application security.

TRAINING

Programming Pathsala | Design And Analysis of Algortihms — Jun' 25 - Jul' 25

- Applied core data structures and algorithms to develop efficient solutions and strengthen problem-solving skills.
- Implemented an LRU Cache using HashMap and Doubly Linked List for O(1) access, insertion, and eviction.
- Built a real-time digital whiteboard using Stack and HashMap for efficient drawing management and undo operations

CERTIFICATES

JNCIA-Junos (Juniper Networks Certified Associate – Junos) | certificate — Sep' 26
Networking Basics | Cisco Networking Academy | certificate — Sep'26
Design and Analysis of Algorithms | Programming Pathsala | certificate — Jul' 25

ACHIEVEMENTS

Solved 500+ coding problems across different platforms. — Jun'26
Awarded Certificate of Volunteering and Community Service by an NGO. — Jun' 24

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology - Computer Science and Engineering | CGPA – 7.78 — Aug' 23 – Present